



Mechanism of molecules crosstalk and the critical role of jasmonic acid on triterpenoid synthesis in *Cyclocarya paliurus* cells under *Aspergillus niger* elicitor

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Abstract

Cyclocarya paliurus (*C. paliurus*) leaves contain multiple health beneficial metabolites, and therefore are often used in functional foods and Chinese herbal medicines. However, breeding difficulties confines its utilization. Consequently, cell suspension cultures were developed to produce the bioactive secondary metabolites of *C. paliurus* leaves, but the content was comparatively low. In the present paper, *Aspergillus niger* elicitor (ANE) was used to stimulate the synthesis of triterpenoids in the suspension cultured *C. paliurus* cells, and the signal molecules crosstalk involved in this elicitation was further studied to interpret the underlying mechanism. Total triterpenoids accumulation in the cultured *C. paliurus* cells elicited by 200 µg/mL ANE was 9.17 times higher than that of the untreated cells. Nitric oxide (NO), hydrogen peroxide (H₂O₂) and jasmonic acid (JA) played important roles in the elicitation as signal molecules. Under the ANE stimulation, the concentrations of NO, H₂O₂ and JA all increased significantly, but presented different profiles and peaked at different time points. Based on series experiments on NO quenching by C-PTIO, H₂O₂ blocking by DMTU, and JA synthesis inhibition by IBU and NDGA, together with exogenous NO, H₂O₂ and JA addition experiments, it was concluded that ANE stimulated triterpenoid biosynthesis in the suspension culture of *C. paliurus* via a complex signal transduction network, in which three deduced and three hypothetical signal transduction pathways might be involved. JA was the critical point in the entire signal network, which connected with NO and H₂O₂ signal pathways. RNA-seq analysis showed that a total of three candidate JA synthesis pathway genes, including one *LOX* and two *OPR* genes, were significantly up-regulated under the ANE stimulation, along with five down-regulated *JAZs* and one up-regulated *JARI* regulating to JA signal transduction, which further validated the pivotal role of JA in ANE elicitation.

Keywords *Cyclocarya paliurus* · *Aspergillus niger* elicitor · Signal transduction · Triterpenoids · Plant cell culture · Elicitation

Abbreviations

SA Salicylic acid
ANE *Aspergillus niger* elicitors
SNP Sodium nitroprusside

DEGs Differentially expressed genes
JA Jasmonic acid
NO Nitric oxide
H₂O₂ Hydrogen peroxide
FPKM Fragment per kilobases of exon model per million mapped reads
FDR False discovery rate
NOS Nitric oxide synthase

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Introduction

Cyclocarya paliurus (Bata1.) Iljinskaja (*C. paliurus*) is a well-known edible and medicinal herb belonging to the family Juglandaceae, which mainly grows in Southern

China (Xie et al. 2008). Its leaves contain multiple kinds of bioactive compounds such as triterpenoids (Sun et al. 2019), flavonoids (Xie et al. 2018), polysaccharides (Xie et al. 2013) and polyphenols (Liu et al. 2018a), and therefore display various bioactivities including anti-tumor (Xuan et al. 2019), regulation of lipid metabolism (Yang et al. 2016) and antidiabetic (Sun et al. 2019). Unfortunately, *C. paliurus* is on the verge of extinction due to difficulties with its reproduction and cultivation, which dramatically limits its use in medicines and foods, and also drives researchers to develop new techniques to better utilize this resources. Plant cell culture technology presents many advantages to the production of plant secondary metabolites, including short production cycle, better controllability, and independence from climate and weather (Hidalgo et al. 2018), and therefore it is regarded as a feasible method to make full use of the *C. paliurus* resources.

A series of studies focused on *C. paliurus* resources utilization had been carried out in our laboratory, such as metabolites identification and quantification (Yin et al. 2017; Li et al. 2013), metabolites bioactivity (Xiong et al. 2018; Hu et al. 2018; Yang et al. 2018), callus induction and screening (Shangguan et al. 2006) and cell suspension culture (Ren et al. 2012; Yin et al. 2013b). A stable cell suspension culture had been established to produce the main bioactive components of *C. paliurus* in our laboratory.

Triterpenoids of *C. paliurus* have attracted attention for its excellent bioactivities (Jiang et al. 2019). Eight triterpenoids, including arjunolic acid, betulinic acid, corosolic acid, maslinic acid, oleanolic acid, ursolic acid, β -amyrin and β -boswellic acid, were identified from the leaves of *C. paliurus* in previous studies (Li et al. 2013). Among them, five had been found in suspension culture of *C. paliurus* (Yin et al. 2013a). To further improve the triterpenoids yield, sodium nitroprusside (SNP) was added into the culture medium as an abiotic elicitor, and the results showed that total triterpenoids yield of the elicited cells was 2.14 times that of the control group (Guo et al. 2014).

Elicitation is one of the most effective ways to stimulate secondary metabolite synthesis in the plant cell, and therefore it is now widely employed to induce novel metabolites or enhance the accumulation of the target metabolites in the plant cell, tissue and organ cultures (Halder et al. 2019). According to chemical properties, elicitors are classified into two types, namely abiotic elicitors (e.g. methyl jasmonate, salicylic acid (SA), AgNO_3 and CaCl_2) and biotic elicitors (e.g. fungal extracts, bacterial extracts and chitin) (Buraphaka and Putalun et al. 2020; El-Bialy and El-Bastawisy et al. 2020). These elicitors can trigger defense or stress-induced responses at a low concentration, which can further regulate the

expression level of some biosynthesis pathway genes via complex signal transduction networks and finally promote the synthesis of secondary metabolites (Halder et al. 2019). Fungal extract is one of the most effective and studied elicitors in the last three decades. Kümritz et al. (2016) reported that adding fungal medium filtrates combined with sucrose feeding significantly enhanced triterpenes content of *Salvia fruticosa* cell suspension cultures by approximately 140%, and the triterpenes yield almost reached the level of intact plants. According to Li et al. 2016a, the glycyrrhetic acid content of the cultured *Glycyrrhiza uralensis* adventitious roots increased by 1.8 times with the addition of *Aspergillus niger* elicitors (ANE). However, the mechanism of ANE elicitation is not well understood, especially the involved signal transduction network.

In the present paper, ANE was used as a biotic elicitor to increase triterpenoids biosynthesis in suspension culture of *C. paliurus*. To interpret the elicitation mechanism, changes in the concentration profile of four signal molecules were determined in detail after ANE induction. The crosstalk among these signal molecules was further studied by a series of experiments to block signal molecules, as well as experiments of exogenous signal molecules addition. Differentially expressed genes (DEGs) associated with jasmonic acid (JA) biosynthesis and transduction were also analyzed to demonstrate the critical role of JA in this elicitation. To the best of our knowledge, at present there is no report on signal molecules crosstalk involved in ANE elicitation in suspension cultured *C. paliurus* cells. Our results and analysis provided references for the further studies on the triterpenoids biosynthesis regulation of the cultured plant cells.

Materials and methods

Cell suspension cultures

According to the method reported by Yin et al. (2017), *C. paliurus* cells (Jiangxi Agricultural University, Nanchang, China) were cultured in a basal liquid MS medium (Murashige and Skoog, medium) supplemented with 2,4-D (2-(2,4-dichlorophenoxy) acetic acid, 0.5 mg/L), NAA (1-Naphthaleneacetic acid, 0.3 mg/L) and KT (6-Furfurylaminopurine, 1.0 mg/L) in the dark at $25 \pm 1^\circ\text{C}$ on a rotary shaker at 115 rpm, and sub-cultured every 6 days.

Biomass determination

Biomass determination was performed in accordance with the method of Yin et al (2013b). The cells of suspension culture were separated from the liquid medium by vacuum extraction filtering, and then dried at 60°C in a vacuum

oven. The dry weight (DW) of the cultured cells were measured until a constant weight was observed.

***Aspergillus niger* elicitor preparation**

Aspergillus niger (CGMCC3.0453) was purchased from China General Microbiological Culture Collection Center (CGMCC). ANE (*Aspergillus niger* elicitor) was prepared according to the method of Xiong et al. (2016).

Elicitation method of ANE on the cultured *C. paliurus* cells

Elicitation was performed in accordance with the method described by Xiong et al. (2016). ANE at a concentration of 200 µg/mL was added into the culture medium on the 4th day after cell inoculation to stimulate the synthesis of triterpenoids. The cultured cells were respectively sampled at 0th, 20th, 40th, 60th and 80th hour after ANE elicitation for triterpenoids and signal molecules determinations.

Triterpenoids extraction and determination

Triterpenoids were extracted by ultrasound (Yin et al. 2013a). A total of 0.5 g fine powder sample was dispersed in 6 ml ethyl acetate and extracted in an ultrasonic extractor at 60 °C for 30 min. After centrifugation at 4000 rpm/min for 10 min, the supernatant was dried at 60 °C, and redissolved in methanol for triterpenoids determination. The triterpenoids determination was carried out according to the HPLC method of Yin et al. (2017). Total triterpenoids content was the sum of maslinic acid, corosolic acid, betulinic acid, oleanolic acid and ursolic acid. Maslinic acid ($\geq 98\%$), corosolic acid ($\geq 98\%$), betulinic acid ($\geq 98\%$), oleanolic acid ($\geq 98\%$) and ursolic acid ($\geq 98\%$) were all purchased from Shanghai Yuanye Bio-Technology Co., Ltd (Shanghai, China).

Signal molecules assay

A total of four signal molecules, including nitric oxide (NO), hydrogen peroxide (H_2O_2), Jasmonic acid (JA) and salicylic acid (SA), were assayed by enzyme-linked immunosorbent assay. NO assay kit was purchased from Beyotime Bio-Technology Co., Ltd (Shanghai, China). H_2O_2 , JA and SA assay kit were all provided by Yihan Bio-Technology Co., Ltd (Shanghai, China). All the assays were performed according to manufacturers' instructions.

Quenching of NO

C-PTIO (Enzo life Sciences, Switzerland), a strong quencher of NO, was used to eliminate NO in the cell

cultures (Xu et al. 2005). Twenty minutes before the addition of ANE, C-PTIO solution was added to the culture medium, and the final C-PTIO concentration was controlled at 0.1, 1 and 100 µM respectively. The content of triterpenoids, H_2O_2 , JA and SA in the cultured cells were respectively determined when the NO was quenched by the added C-PTIO.

Blocking of H_2O_2

DMTU (Sigma-Aldrich, USA) was used as H_2O_2 scavenger (Lin et al. 2005). 20 min before the ANE elicitation, DMTU (final concentration of 0.5, 1 and 2 mM) was added into the culture medium. The changes of triterpenoids accumulation were tested as well as the concentration changes of NO, JA and SA content when H_2O_2 was blocked by DMTU.

Inhibition of JA synthesis

Two inhibitors, namely ibuprofen (IBU) and nordihydroguaiaretic (NDGA), were simultaneously used to block JA synthesis in the cultured *C. paliurus* cells (Nojiri et al. 1996; Menke et al. 1999). Three combinations of IBU and NDGA concentration were applied into the culture medium 20 min before the ANE elicitation, including 25 µM IBU and 25 µM NDGA, 50 µM IBU and 50 µM NDGA, and 100 µM IBU and 100 µM NDGA, which was used to investigate the role of JA in the triterpenoid synthesis stimulation of ANE.

Exogenous addition of signal molecules

Sodium nitroprusside (SNP, Sigma-Aldrich, USA, 99%) was used as exogenous NO donor to activate the NO signal pathway (Bethke et al. 2006). Methyl jasmonate (MJA, Sigma-Aldrich, USA, 95%) was applied as JA donor to generate JA signal molecules (Creelman et al. 1992). SNP, MJA and H_2O_2 (Xilong Chemical Co., Ltd. Guangdong, China, 30%) were respectively added into the *C. paliurus* cell culture medium at the same time of ANE elicitation as positive signal molecules control group, whose final concentrations were set at 150 µM, 10 µM and 50 µM respectively.

RNA extraction and sequencing

The treated and untreated (control) cells were harvested 20 h after ANE elicitation and immediately frozen in liquid nitrogen and stored at -80 °C for total RNA extraction. Total RNA was extracted from the cultured *C. paliurus* cells using plant RNA purification reagent (Invitrogen, Carlsbad, CA, USA). The quantity and quality of total

RNA were determined by Nanodrop 2000 (Thermo Fisher, America) and Agilent 2100. Poly (A) mRNA (Agilent 2100 Bioanalyzer System) was enriched from total RNA using Oligo (dT) magnetic beads. The enriched mRNA was randomly shared into short fragments and then used to synthesize the cDNA using reverse transcriptase and random primers. These synthesized cDNA fragments were purified and dissolved in EB buffer for end repairing and adding of poly (A), and then connected with sequencing adapters. Suitable fragments were selected for PCR amplification as templates to establish cDNA library. Finally the cDNA library was sequenced using an Illumina HiSeq 4000 platform with the 2×151 bp paired-end reads.

Reads alignment and de novo assembly

The raw reads were filtered by removing adaptor sequences, ambiguous reads and low quality reads (quality value < 20). Processed reads were assembled into transcript contigs using Trinity software (<http://trinityrnaseq.sourceforge.net/>, release-20140413), to yield unigenes.

Functional annotation and classification of unigenes

All assembled unigenes were annotated using BLASTx alignment (e-value < 0.00001) against the following publicly available protein databases: NCBI non-redundant (NR), Swiss-Prot, Pfam, eukaryotic orthologous groups (COG) and KEGG. Combining NR annotation with the Blast2GO program (Conesa et al. 2005), GO annotations of the unigenes were obtained. All GO functional classifications were produced by WEGO software (Ye et al. 2006). Pathway assignments were performed in conjunction with the KEGG database. When a conflict occurred among the database results, a priority order of alignments was NR, Swiss-Port, KEGG, GO and COG.

Identification of differentially expressed genes

Normalized expression levels of the unigenes were calculated as fragment per kilobases of exon model per million mapped reads (FPKM) by the RSEM software (version number: 1.1) (Mortazavi et al. 2008). The differential expression analysis of any two groups was analyzed using the EdgeR software (version number: 2.1.2) and false discovery rate (FDR) method (Reiner et al. 2003). Significantly differentially expressed genes (DEGs) were identified by the threshold of $FDR < 0.05$ and $\log_2|\text{Fold Change}| \geq 1$ (Audic and Claverie, 1997). To further identify significantly over-represented metabolic pathways or signal transduction pathways, all DEGs were mapped to

the KEGG database with a threshold E-value of 10^{-5} for pathway enrichment analysis.

RT-qPCR determination and analysis

To generate a comprehensive information of *C. paliurus* transcriptome, total RNA was isolated from *C. paliurus* cell elicited by ANE for 0 and 20 h respectively using TRIzol® Reagent (Invitrogen) according to the manufacturer's instructions. The concentration and purity of the RNA were determined by agarose gel electrophoresis method. After treatment of DNase I (Sangon, China), 4 µg of RNA was used in reverse transcription with RevertAid Premium Reverse Transcriptase kits (Thermo Scientific™, China). RT-qPCR was performed using SG Fast qPCR Master Mix (ABI, USA) and ran on Step One Plus Real-Time PCR Detection System (ABI, USA). The cycling conditions were initial denaturation at 95 °C for 3 min, followed by 45 cycles of 95 °C for 7 s, 57 °C for 10 s, and 72 °C for 15 s. All DEGs determinations were triplicated with Actin as reference genes. Gene-specific primer pairs (Table S1) were designed by Primer 5.0 software. Relative expression levels for each unigene were compared between the two samples with the $2^{-\Delta\Delta C_t}$ method (Livak and Schmittgen, 2001).

Data analysis

All the experiments were performed in three biological replicates and the results were showed as the mean $\pm$ standard deviation. The data was analyzed by SPSS software with Duncan's multiple range test (DMRT) (at $P \leq 0.05$).

Results

Stimulation effects of ANE on triterpenoids synthesis in cultured cells

To further enhance the triterpenoids production, ANE, a widely used biotic elicitor, was applied to stimulate triterpenoids synthesis in cultured cells. Results (Fig. 1) indicated that the accumulation of triterpenoids in the cultured *C. paliurus* cells was significantly enhanced under the elicitation of ANE at a concentration of 200 µg/mL, while the biomass was unaffected. The yield of total triterpenoids peaked 60 h after ANE addition. The peak total triterpenoids yield of the elicited cells was 9.17 times higher than that of the untreated cells. However, the elicitation mechanism of ANE was still unclear. To better understand the mechanism, crosstalk between four signal molecules was further investigated using RNA-seq.

Profile changes of four signal molecules concentration under the ANE elicitation

After elicitation with 200 $\mu\text{g/mL}$ ANE, the concentrations of NO, H_2O_2 and JA all increased significantly in the cultured *C. paliurus* cells, but presented different change profiles and peaked at different times (as shown in Fig. 2a–c). While, SA concentration fell gradually at the beginning, then kept relatively stable and lower than that of the untreated cells until 80 h post-elicitation (Fig. 2d). Consequently, it was deduced that NO, H_2O_2 and JA were involved in signal transduction after ANE elicitation.

A rise in NO concentration is a common physiological reaction in plant cells when stimulated by elicitors (Xu

et al. 2005). Our results (Fig. 2a) showed that NO concentration increased swiftly after the addition of ANE, and peaked at 15 min (significantly higher than that of the untreated cells), then decreased sharply and maintained at a similar level of the untreated cells from 4 to 80 h post-elicitation. Similarly, H_2O_2 concentration also increased rapidly under the ANE stimulation, and also reached a high concentration at 15 min (Fig. 2b). Nevertheless, unlike NO, H_2O_2 concentration remained at a high level from 15 min to 80 h, which was significantly higher than that in the untreated cells (Fig. 2b). The concentration changes of JA lagged behind that of NO and H_2O_2 (Fig. 2a–c). Within 6 h after ANE elicitation, JA concentration was almost the same as that in un-elicited cells. From the 7th to 80th h post-elicitation, JA concentration was significantly higher than that of the control group, with two peaks appearing at the 18th and 42nd h after ANE addition respectively (Fig. 2c). The highest concentration of JA occurred at the 42 h post-elicitation. In view of the time sequence changes of these signal molecules, a series of experiments of NO quenching by C-PTIO, H_2O_2 blocking by DMTU, and JA synthesis inhibition by IBU and NDGA were further performed to understand the crosstalk among these signal molecules, together with exogenous NO, H_2O_2 and JA addition experiments.

Changes caused by NO quenching and exogenous addition

NO concentration decreased gradually with the C-PTIO concentration increasing from 0.1 to 100 $\mu\text{mol/L}$, and decreased to the level of the untreated cells at 100 $\mu\text{mol/L}$, but the C-PTIO addition and NO change had no influence on the cell growth (biomass was given in Fig. 3a, b). When 100 $\mu\text{mol/L}$ C-PTIO was applied to the cultured cells elicited by 200 $\mu\text{g/mL}$ ANE, H_2O_2 concentration was approximately the same as that in the elicited cells, while JA concentration declined significantly, but still higher than that of the control group (neither ANE nor C-PTIO was added) (Fig. 3c, d). With NO blocked by C-PTIO, triterpenoids yield of the ANE elicited cells reduced by 21.28%, but its concentration was still markedly higher than that of the control group (Fig. 3e).

To further interpret the role of NO and its relationship with H_2O_2 and JA, 150 $\mu\text{mol/L}$ SNP (exogenous NO donor) was added into the culture medium of *C. paliurus* cells. Results (Fig. 3f, g) indicated that exogenous NO showed no effect on H_2O_2 concentration, while raised the concentration of JA significantly. However, JA concentration in the cells treated by exogenous NO was still lower than that in ANE elicited cells. Under the exogenous NO stimulation, triterpenoids yield was 4.86 times that of the

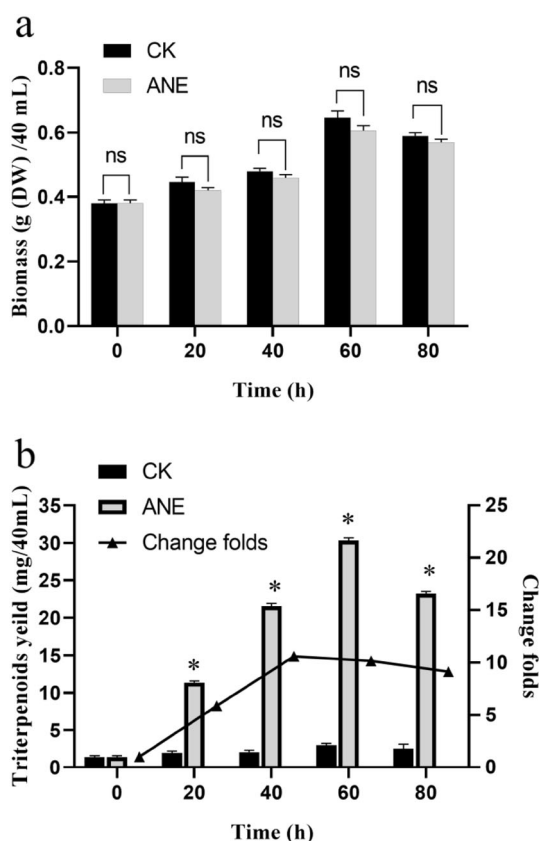


Fig. 1 Comparison of biomass (a) and triterpenoids yield (b) between ANE elicited and un-elicited *C. paliurus* cells. Suspension culture conditions: *C. paliurus* cells were cultured in MS medium supplemented with combined hormones (1.0 mg/L KT + 0.5 mg/L 2, 4-D + 0.3 mg/L NAA), 6 g/L agar and 30 g/L sucrose, and cultivated in the dark at $25 \pm 1^\circ\text{C}$ on a rotary shaker at 115 rpm. ANE elicitation parameters: ANE at a concentration of 200 $\mu\text{g/mL}$ was added into the culture medium on the 4th day after cell inoculation; cells were respectively harvested at 0, 20, 40, 60 and 80 h after elicitation by ANE. CK: the control group cells; ANE: The cultured cells elicited by ANE. Significant difference analysis was conducted by unpaired t-test with a *p*-value of at least less than 0.05; “NS” meant not significant. Superscript symbol “*” indicates that the changed significantly

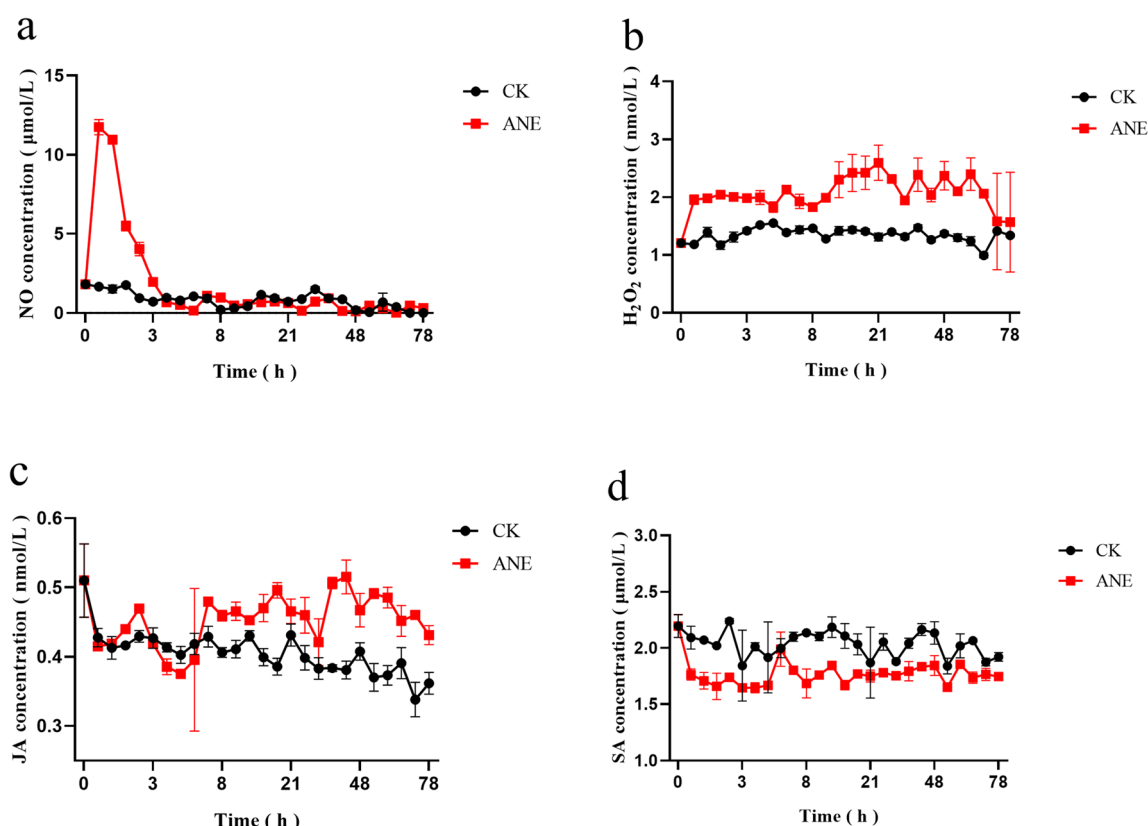


Fig. 2 The change profiles of NO (a), H₂O₂ (b), JA(c) and SA (d) concentration of suspension cultured *C. paliurus* cells at different times after ANE elicitation. Suspension culture conditions: *C. paliurus* cells were cultured in MS medium supplemented with combined hormones (1.0 mg/L KT + 0.5 mg/L 2, 4-D + 0.3 mg/L

NAA), 6 g/L agar and 30 g/L sucrose, and cultivated in the dark at 25 ± 1 °C on a rotary shaker at 115 rpm. ANE (200 μg/mL) was added into the culture medium on the 4th day after cell inoculation. CK: the un-elicited cells; ANE: The cultured cells elicited by ANE

control group, but significantly lower than that of the ANE treated cells (Fig. 3h).

Based on the above analysis, the following three conclusions were obtained: firstly, ANE elicitation promoted the synthesis of triterpenoids in *C. paliurus* cells partly through NO signal pathway; secondly, JA was involved in NO pathway in ANE elicitation, and located in downstream of NO pathway; thirdly, H₂O₂ synthesis might be independent on the change of NO concentration.

Changes caused by H₂O₂ blocking and exogenous addition

As DMTU concentration rose from 0.5 to 2 mmol/L, the elimination rate of H₂O₂ increased in the *C. paliurus* cell cultures accordingly (Fig. 4a). After treatment with of 2 mmol/L DMTU, H₂O₂ concentration in the ANE

elicitation cells decreased to the level of the untreated cells. Meanwhile JA concentration was reduced by 14.53%, but it was still higher than that in the control (Fig. 4c). 2 mmol/L DMTU showed no significant impact on cell growth (Fig. 4b), but decreased the triterpenoids yield of the ANE treated cells by 13.76% (Fig. 4d). Although triterpenoids accumulation in ANE elicited cells was reduced by 2 mmol/L DMTU, it was still higher than that in the control group (Fig. 4d).

When exogenous 50 μmol/L H₂O₂ was added into the culture medium of *C. paliurus* cells, JA concentration increased by 11.31%, but it was still significantly lower than that of the ANE elicited cells (Fig. 4e). Stimulated by 50 μmol/L H₂O₂, triterpenoids accumulation rose by 78.35%, but it was notably lower than that in the ANE treated cells (Fig. 4f).

Fig. 3 Change profiles of NO, H_2O_2 and JA content, biomass, and triterpenoids yield of cultured *C. paliurus* cells caused by NO quenching and exogenous addition. **a**: the contents of NO in cultured *C. paliurus* cells under different concentrations of C-PTIO; **b** biomass of the cultured *C. paliurus* cells under different concentrations of C-PTIO; **c** the contents of H_2O_2 in the ANE elicited *C. paliurus* cells with or without the addition of the 100 $\mu\text{mol/L}$ C-PTIO; **d** the contents of JA in the ANE elicited *C. paliurus* cells with or without the addition of the 100 $\mu\text{mol/L}$ C-PTIO; **e** the yield of triterpenoids in the ANE elicited *C. paliurus* cells with or without the addition of the 100 $\mu\text{mol/L}$ C-PTIO; **f** the contents of H_2O_2 in the ANE or SNP (adding 150 $\mu\text{mol/L}$ SNP, exogenous NO donor) elicited *C. paliurus* cells; **g** the contents of JA in the ANE or SNP elicited *C. paliurus* cells; **h** the yield of triterpenoids in the ANE, ANE combined with 100 $\mu\text{mol/L}$ C-PTIO, and SNP treated *C. paliurus* cells. CK: the control group cells; ANE: the cultured cells elicited by ANE; ANE + C-PTIO: the cultured cells adding ANE and C-PTIO; SNP: the cultured cells adding to SNP; ANE + C-PTIO + SNP: the cultured cells adding to ANE, C-PTIO and SNP. Different superscript lowercase letters “a–d” indicates a significant difference between data at $p < 0.05$. Statistical analysis was performed with ANOVA followed by Duncan’s multiple range test (DMRT)

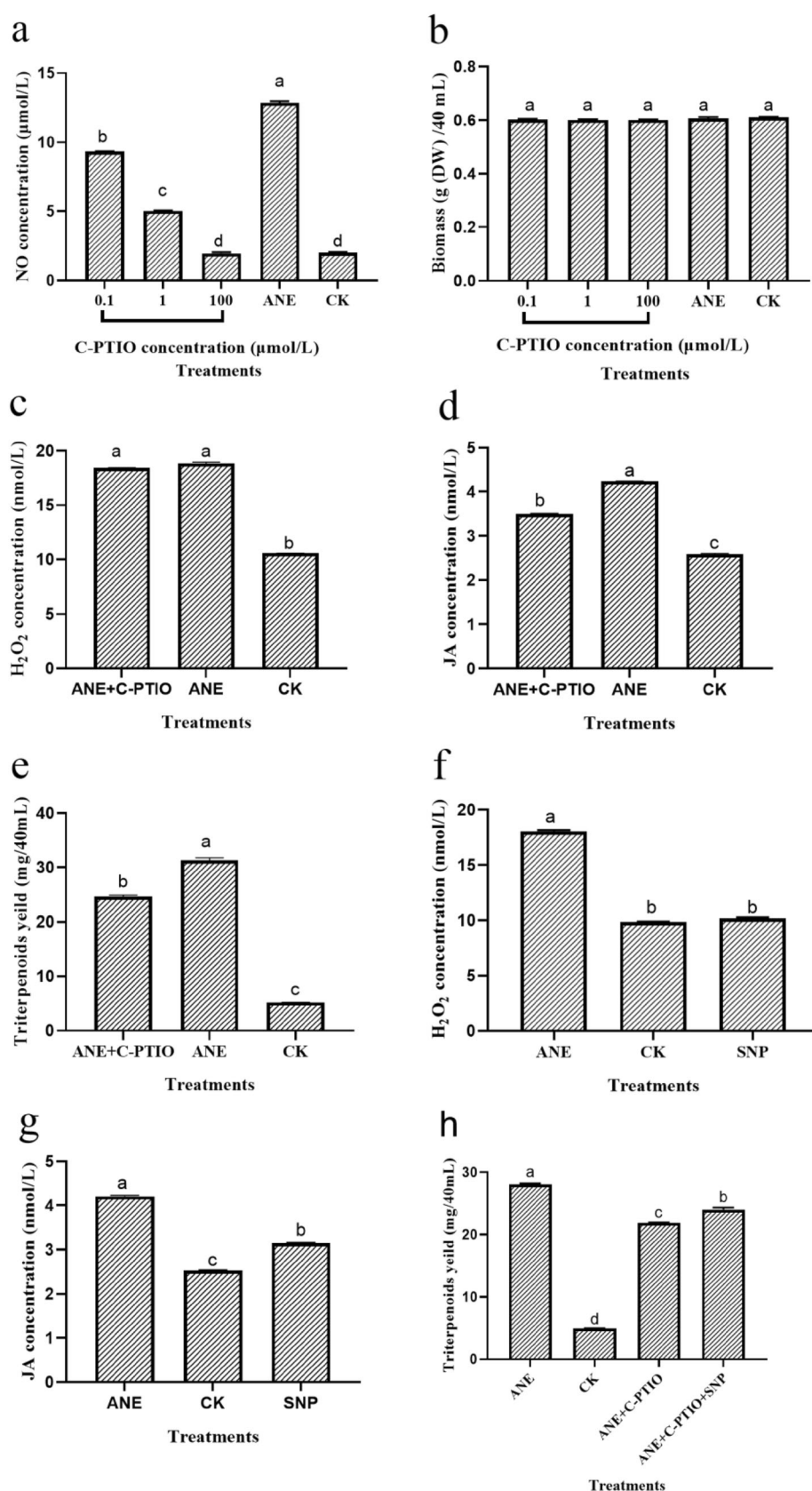
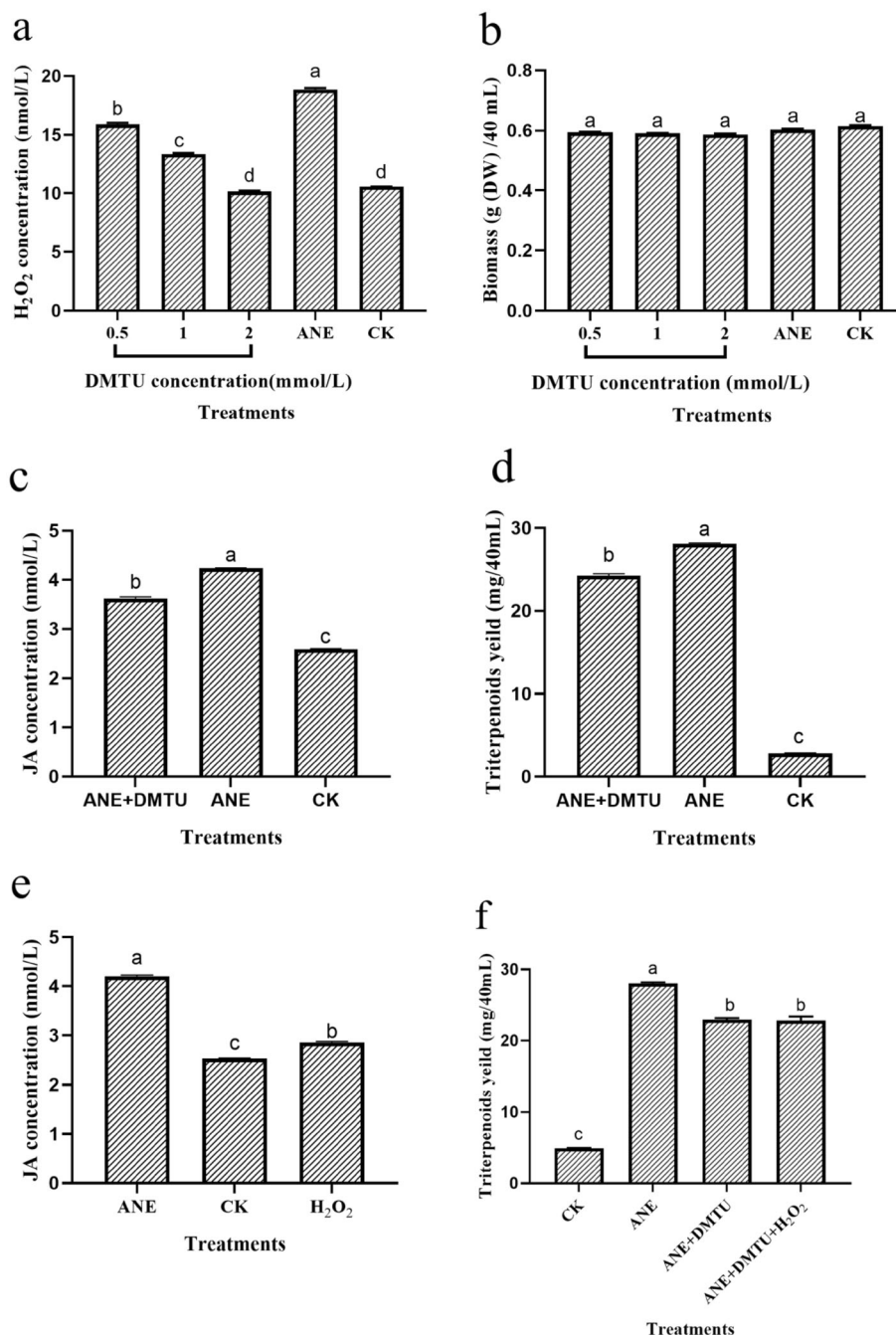


Fig. 4 Change profiles of H_2O_2 and JA content, biomass, and triterpenoids yield of cultured *C. paliurus* cells caused by H_2O_2 blocking and exogenous addition. **a** the contents of H_2O_2 in cultured *C. paliurus* cells under different concentrations of DMTU; **b** biomass of the cultured *C. paliurus* cells under different concentrations of DMTU; **c** the contents of JA in the ANE elicited *C. paliurus* cells with or without the addition of the 2 mmol/L DMTU; **d** the yield of triterpenoids in the ANE elicited *C. paliurus* cells with or without the addition of the 2 mmol/L DMTU; **e** the contents of JA in the ANE, ANE combined with 2 mmol/L DMTU and 50 μ mol/L H_2O_2 elicited *C. paliurus* cells; **f** the yield of triterpenoids in the ANE, ANE combined with 2 mmol/L DMTU, and 50 μ mol/L H_2O_2 treated *C. paliurus* cells. CK: the control group cells; ANE: the cultured cells elicited by ANE; ANE + DMTU: the cultured cells adding ANE and DMTU; H_2O_2 : the cultured cells adding H_2O_2 ; ANE + DMTU + H_2O_2 : the cultured cells adding ANE and DMTU. Different superscript lowercase letters “a–d” indicates a significant difference between data at $p < 0.05$. Statistical analysis was performed with ANOVA followed by Duncan’s multiple range test (DMRT)



In “**Conclusion**”, ANE elicitation improved triterpenoids synthesis in *C. paliurus* cells partly through H_2O_2 signal pathway, and the change of H_2O_2 concentration could cause the change of JA concentration and finally led to a fluctuation of triterpenoids yield accordingly.

Changes caused by JA synthesis inhibition and exogenous JA addition

The combined application of IBU and NDGA could effectively inhibit JA synthesis in the cultured *C. paliurus*

cells. With the addition of 100 μ mol/L IBU and NDGA, JA concentration in the ANE elicited cells declined to the level of the un-elicited cells (Fig. 5a), while cell growth was unaffected (Fig. 5b). Under the inhibition of 100 μ mol/L IBU and NDGA, triterpenoids yield was reduced to 417.25 mg/L, which was 59.41% of the ANE treated cells, but still significantly higher than that of the control (Fig. 5c). When 10 μ mol/L MJA was used as JA donor in the culture medium, triterpenoids yield increased to 608.25 mg/L, which was 4.92-fold change higher than untreated cells (Fig. 5d).

Comparatively speaking, blocking JA resulted in the lowest of yield of triterpenoids in the ANE elicited *C. paliurus* cells (Fig. 5c), followed by blocking NO and H₂O₂ (Figs. 3e and 4d). Similarly, exogenously fed JA lead to the highest increase of triterpenoids yield in the cultured *C. paliurus* cells (Fig. 5d), followed by NO and H₂O₂ (Figs. 3h and 4f).

Consequently, it was deduced that JA was the critical signal molecule for triterpenoids yield promotion in the ANE elicitation, and the change of JA concentration could cause a notable change of triterpenoids accumulation in the culture *C. paliurus* cells.

Summary of Signal molecules crosstalk and presumed signal transduction pathways involved in ANE elicitation

Our signal molecules blocking and exogenous addition experiments suggested that NO, H₂O₂ and JA were all involved in the responses of ANE elicitation in the suspension culture of *C. paliurus*; blocking NO, H₂O₂ and JA lead to a decrease in triterpenoids accumulation in the ANE elicited cells. In contrast, the exogenous addition of NO, H₂O₂ and JA resulted in the accumulation of triterpenoids, although their increase was lower than that of ANE; both NO and H₂O₂ had apparent impacts on JA synthesis; the change of NO concentration showed no effect on the H₂O₂

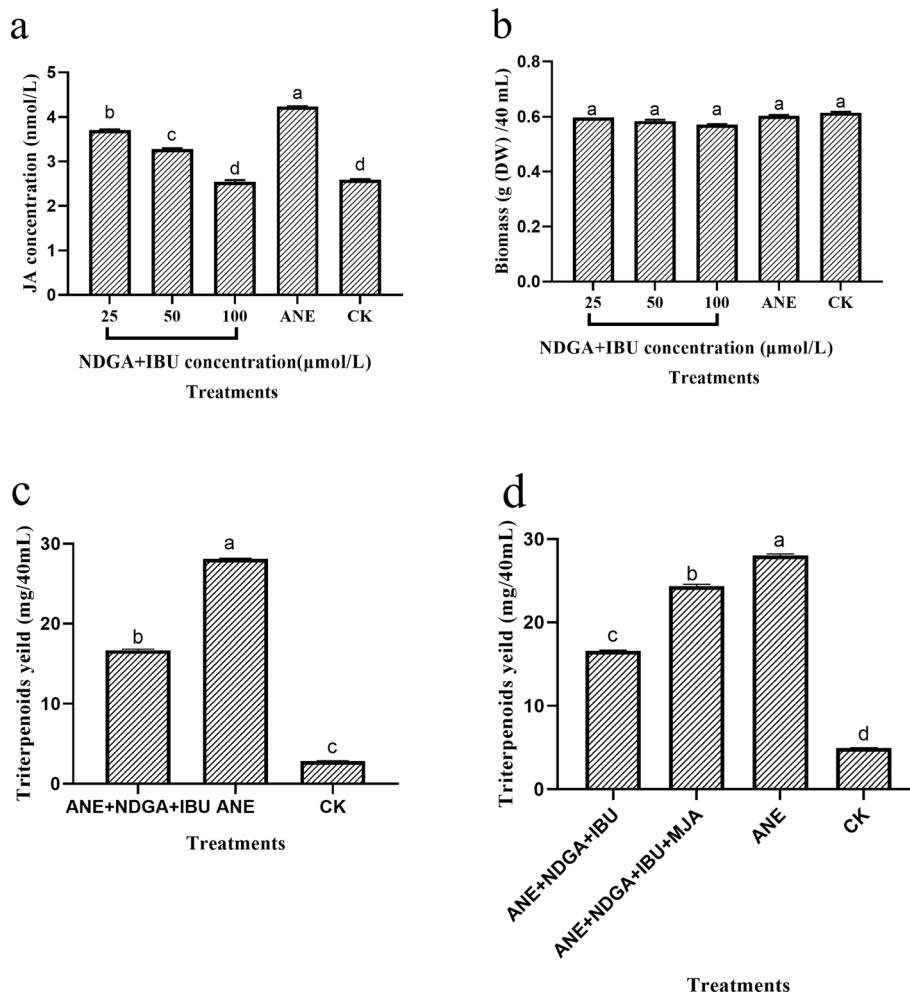


Fig. 5 Change profiles of JA content, biomass, and triterpenoids yield of cultured *C. paliurus* cells causing by JA inhibiting and exogenous addition. **a** the contents of JA in cultured *C. paliurus* cells under different concentrations of NDGA + IBU; **b** biomass of the cultured *C. paliurus* cells under different concentrations of NDGA + IBU; **c** the yield of triterpenoids in the ANE elicited *C. paliurus* cells with or without the addition of the 100 μmol/L NDGA + IBU; **d** the yield of triterpenoids in the ANE, ANE combined with 100 μmol/L NDGA + IBU, and MJA (adding 10 μmol/L MJA, exogenous JA

donor) treated *C. paliurus* cells. CK: the control group cells; ANE: the cultured cells elicited by ANE; ANE + NDJA + IBU: the cultured cells adding ANE and NDJA + IBU; ANE + NDJA + IBU + MJA: the cultured cells adding ANE, NDJA + IBU and MJA. Different superscript lowercase letters “a–d” indicates a significant difference between data at $p < 0.05$. Statistical analysis was performed with ANOVA followed by Duncan’s multiple range test (DMRT)

synthesis; the influence of JA on triterpenoids biosynthesis was higher than that of NO and H₂O₂. Based on the above analyses, three signal transduction pathways were deduced, which were shown in Fig. 6 and described as following: (1) ANE → receptor → nitric oxide synthase (NOS) → NO → jasmonic acid synthesis enzymes → JA → TFs → triterpenoids synthesis enzymes → triterpenoids; (2) ANE → receptor → NADPH oxidase → H₂O₂ → jasmonic acid synthesis enzymes → JA → TFs → triterpenoids synthesis enzymes → triterpenoids; (3) ANE → receptor → NADPH oxidase → H₂O₂ → nitric oxide synthase (NOS) → NO → jasmonic acid synthesis enzymes → JA → TFs → triterpenoids synthesis enzymes → triterpenoids. In addition, there might be another three hypothetical signal transduction pathways involved in ANE elicitation, which need to be further validated: (1) ANE → receptor → nitric oxide synthase (NOS) → NO → TFs → triterpenoids synthesis enzymes → triterpenoids; (2) ANE → receptor → NADPH oxidase → H₂O₂ → TFs → triterpenoids synthesis enzymes → triterpenoids; (3) ANE → receptor → other pathways independent of NO, H₂O₂ and JA → TFs → triterpenoids synthesis enzymes → triterpenoids. In summary, ANE elicitation improved the triterpenoids accumulation in the suspension cultured *C. paliurus* cells via a complex signal transduction network, in which NO, H₂O₂ and JA were involved, and JA was the critical signal molecule.

RNA-seq and de novo assembly

Three cDNA libraries from the cultured *C. paliurus* cells, named CK (control), E_{20h} (cells elicited for 20 h), E_{60h} (cells elicited for 60 h) respectively, were sequenced using the Illumina Hiseq 4000 platform, which generated 65,074,242, 66,016,592 and 53,951,932 raw reads, respectively (Table S2). After data filtering and stringent quality evaluation, clean reads were obtained. Subsequently, these clean reads were assembled into 88,144 transcripts with an N50 of 1,590 bp and an average length of 869 bp, which were then joined into 67,230 unigenes with an N50 of 1375 bp and the average length of 744 bp (Table S3). Furthermore, the length and number of

transcripts and unigenes were statistically analyzed. The length of transcripts and unigenes ranged from 201 to 17,098 bp and the majority of them distributed in 201–400 bp range, accounting for 45.56% and 52.99% respectively (Fig. S1). The above-mentioned data indicated that the generated unigenes in our experiments were of good quality, and therefore suitable for further annotation.

Functional annotation and classification of unigenes

A total of 33,470 unigenes (accounting for 49.78% of all unigenes) were successfully annotated against seven public databases, namely NR, Swiss-Prot, KOG KEGG, GO, COG (Table S4). A total of 3972 annotated unigenes of *C. paliurus* showed homology to sequences from *Vitis vinifera*, followed by the *Theobroma cacao* (3417 unigenes), *Prunus persica* (2702 unigenes), *Prunus mume* (2548 unigenes) and *Morus notabilis* (1807 unigenes) (Fig. S2).

All *C. paliurus* unigenes were aligned to the COG database for prediction and classification by possible function. Overall, a total of 7,694 unigenes (11.44%) were classified into 25 COG functional categories (Fig. S3 a). Our results indicated that 1,008 annotated unigenes (13.10%) belong to the “General function prediction only” group, which was the largest among the 25 COG functional categories, followed by “Signal transduction mechanisms” (966, 10.11%). No unigenes were classified as “Extracellular structures” and “Nuclear structure”.

According to the annotation against Gene Ontology (GO), altogether 19,398 unigenes were classified into three main categories: “cellular component” (42,253 unigenes, 32.91%), “molecular function” (24,703 unigenes, 19.24%), and “biological process” (61,440 unigenes, 47.85%), which were further subdivided into 65 sub-categories. (Fig. S3 b). Within the cellular component category, cell sub-category (8,221 unigenes, 19.46%) and cell part sub-category (8,221 unigenes, 19.46%) were the two most significant representation. Among the molecular function unigenes, the majority were classified into “binding sub-category” (11,100 unigenes, 44.93%) and “catalytic activity sub-category” (10,098 unigenes, 40.88%).

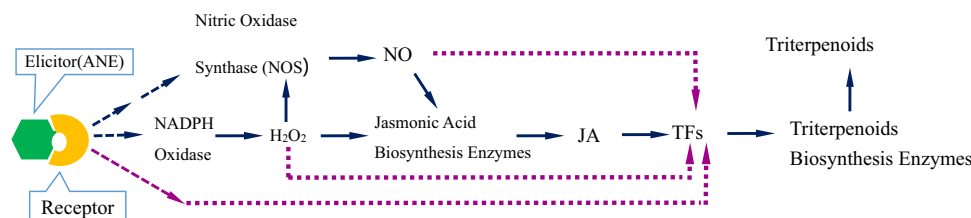


Fig. 6 The presumed signal transduction network of ANE elicitation in suspension cultured *C. paliurus* cells. Blue arrow line (including solid line and dotted line): signal transduction pathway deduced

according to the resultant analyses in the present paper; purple dotted arrow line: hypothetical signal transduction pathway

To better understand the biological functions of these unigenes in the cultured *C. paliurus* cells, a total of 13,634 unigenes (20.28%) were mapped to 348 KEGG pathways and divided into five branches including Metabolism; Genetic Information Processing, Environmental Information Processing, Cellular Processes and Organismal Systems (Fig. S3c). The most represented pathways were Metabolic pathways (2778 unigenes, 18.74%), followed by biosynthesis of secondary metabolites (1336 unigenes, 9.80%). In addition, 29 metabolic pathways played a role in the growth and metabolism of cultured cells (Table S5), including phenylpropanoid biosynthesis (179 unigenes), terpenoid backbone biosynthesis (75 unigenes) and flavonoid biosynthesis (42 unigenes).

Overall analysis of differentially expressed genes

From the perspective of differentially expressed genes (DEGs), ANE elicitation caused great changes in metabolism related genes expression (Fig. S4), which might further result in markedly changes regarding metabolite synthesis and accumulation in cultured *C. paliurus* cells. A total of 24,788 DEGs were found between the un-elicited cells and elicited cells treated for 20 h, which consisted of 12,699 up-regulated DEGs and 12,089 down-regulated DEGs. Among these DEGs, 774 genes were significantly up-regulated, while 876 were significantly down-regulated. Similarly, between the un-elicited cells and elicited cells treated for 60 h, there were 340 significantly up-regulated genes and 129 significantly down-regulated genes. Furthermore, cells elicited for 20 h showed a different gene expression profile when compared 60 h. Between these two treated groups, 406 significant up-regulation and 224 significant down-regulation genes were identified. The three groups of DEGs were further analyzed. Among them, 1650, 469 and 630 genes were significantly differentially expressed, and 35 genes belonged to the differentially expressed genes in three samples (Fig. S5).

Analysis of DEGs involved in the JA synthesis pathway under ANE treatment

To further validate the key role of JA in the responses of ANE elicitation in *C. paliurus* cells, the expression levels of JA biosynthetic genes such as *LOX*, *AOS*, *AOC*, and *OPR* were detected using comparative transcriptome sequencing. A total of seven annotated candidate *LOX* unigenes were up-regulated, among which one was significantly up-regulated (Table 1). Within the six up-regulated *OPR* unigenes, two were significantly up-regulated. The expression of two *AOC* and one *AOS* unigenes increased, but these increases were not statistically significant (Table 1). The above DEGs analysis associated with

JA synthesis was in consistence with the determination data of JA concentration in the elicited cells. Our results indicated that ANE elicitation activated the expression of JA synthesis pathway genes, which finally enhanced the triterpenoids accumulation in the culture cells.

Analysis of DEGs involved in the JA signal transduction pathway under ANE treatment

Previous studies showed that JAZ and JAR1 were key transcription factors in the JA signal transduction pathway, and played critical roles in the accumulation of secondary metabolites (Afrin et al. 2015; Staswick et al. 2004). In the present study, a total of seven JAZ candidate unigenes were annotated, among which five were down-regulated, one remained almost unchanged, while one was up-regulated (Table 2). In addition, one JAR1 candidate unigene (c39500_g1) was annotated and observed to be up-regulated, whose FPKM increased from 48.614 to 77.295 at 20 h after ANE elicitation (Table 2). Consequently, it was speculated that the down-regulation of JAZs together with the up-regulation of JAR1 mediated ANE elicitation signal transduction, and led to the rising of triterpenoids synthesis in the cultured cells, which further validated the above deduced JA signal pathway and the critical role of JA in ANE elicitation.

Validation of DEG expression levels by qRT-PCR

A total of 7 candidate DEGs, including four JA synthesis pathway genes (*LOX*, *AOS*, *AOC*, and *OPR*), three JA signal transduction pathway TFs (*JAR1*, *JAZ*, and *MYC2*) were further validated by RT-qPCR. The results (Fig. 7) showed that the expression of all candidate DEGs were accordant with the analysis results obtained from RNA-seq, which indicated that our RNA-seq data were reliable.

Discussion

Fungal elicitors, as highly effective stimulators, can trigger rapid defense responses when applied to in vitro plant cell, tissue and organ cultures, and thus stimulate secondary metabolites biosynthesis by activating the expression of specific genes (Liang et al. 2018; Li et al. 2016b). Our elicitation experiment suggested that ANE can significantly improve the triterpenoids yield of the cultured *C. paliurus* cells, and therefore has promising application prospects in the production of plant-derived triterpenoids. However, more investigations are needed to better understand the elicitation mechanism. Our previous report showed that ANE elicitation up-regulated the expression of triterpenoid synthesis pathway genes, and therefore enhanced

Table 1 Summary of differentially expressed genes in JA synthesis pathway of suspension culture of *C. paliurus*

Unigene name	Unigene code	FPKM of CK	FPKM of E _{20h}	Trends
LOX	c41476_g2	32.342	36.76	Up
	c33483_g1	16.031	16.534	Up
	c41476_g3	74.344	329.943	Up*
	c17385_g1	22.275	25.741	Up
	c33483_g1	16.031	16.534	Up
	C45031_g1	17.787	22.979	Up
	c17385_g1	22.275	25.741	Up
AOS	c43559_g1	0.618	2.196	Up
AOC	c10965_g1	38.972	41.961	Up
	c35833_g1	58.4	68.947	Up
OPR	c24740_g1	8.136	16.473	Up
	c24761_g1	3.648	4.199	Up
	c32622_g1	19.969	163.079	Up*
	c36002_g1	648.51	853.441	Up
	c36002_g2	519.983	1052.36	Up
	c39431_g1	160.52	584.483	Up*

LOX: lipoxygenase; AOS: allene oxide synthase; AOC: allene oxide cyclase; OPR: oxo-phytodienoic acid reductase

CK: the control group cells; E_{20h}: The cultured cells elicited by ANE for 20 h

Superscript symbol “*” indicates that this unigene expression changed significantly

Table 2 Summary of differentially expressed genes in JA signal transduction pathway of suspension culture of *C. paliurus*

Unigene name	Unigene code	FPKM of CK	FPKM of E _{20h}	Trends
JAR1	c39500_g1	48.614	77.295	Up
JAZ	c27861_g1	10.327	4.938	Down
	c33284_g1	2.664	5.504	Up
	c38204_g1	9.757	8.196	Down
	c44879_g3	24.427	24.881	Up
	c44879_g4	58.892	32.683	Down
	c45306_g2	38.711	31.479	Down
	c45306_g3	44.28	26.045	Down
MYC2	c29186_g1	27.69	21.089	Down
	c37338_g1	16.89	14.904	Down
	c43784_g1	103.877	85.936	Down

JAR1: JA-amino acid synthetase 1; JAZ: jasmonate ZIM domain-containing protein; MYC2: myelocytomatosis protein 2

CK: the control group cells; E_{20h}: The cultured cells elicited by ANE for 20 h

triterpenoid accumulation in cultured *C. paliurus* cells (Liu et al. 2016), but the signaling mechanism involved in up-regulation of triterpenoid gene expression has not yet been elucidated.

Previous studies indicated that there are four common signal molecules, namely NO, H₂O₂, JA and SA, which mediate signal transduction when plants face various environmental stresses such as fungal invasion, insect attack, drought and osmotic pressure (Saxena et al. 2016; Yan et al. 2020; Zhang and Li et al. 2019). Usually, these signal molecules play their roles in the complicated signal

transduction network harmoniously, and result in differential expression of some related key genes, which further causes various physiological responses in plants (Lv et al. 2019; Astier et al. 2016).

Signal transduction is a very complex biological process, and therefore has attracted much attention in recent decades. Domingos et al. (2015) comprehensively reviewed the characteristics, signal transduction pathways and mechanisms, and functions of NO as a plant signaling molecule, and pointed out that NO played an important role in responses of plants facing various kinds of biotic and

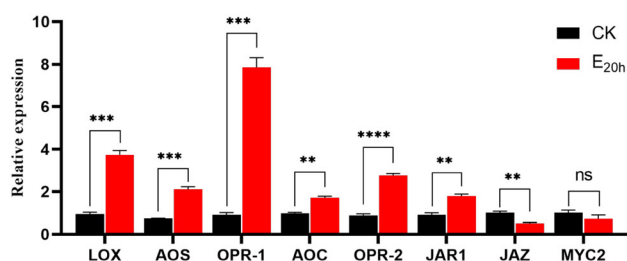


Fig. 7 RT-qPCR determination results of the candidate differentially expressed genes between the ANE treated *C. paliurus* cells for 0 and 20 h. Data were described as Mean $\pm$ SD ($n = 3$). The cell samples elicited by ANE for 0, and 20 h were designated as “CK”, and “E_{20h}”, respectively. Significant difference analysis was conducted by unpaired t-test with a p -value of at least less than 0.05; “NS” meant not significant. Superscript symbol “*” indicates that the changed significantly

abiotic stress. Turner et al. (2002) elaborated on the JA signaling pathway, as well as the biosynthesis and regulation, signaling functions, and relationship of JA with other signal molecules like SA and ethylene. Although extensively studied, signal transduction mechanism such as signal molecules crosstalk have not been fully understood yet, because numerous highly complicated pathways involving various signal molecules and related proteins are involved.

Xu and Dong. (2006a) suggested that NO mediated the fungal elicitor-induced taxol biosynthesis in the *T. chinensis* suspension cultured cells through both reactive oxygen species-dependent and -independent signal pathways. They showed that both NO quenching and blocking of reactive oxygen species (ROS, including H₂O₂) inhibited elicitor-induced taxol production, which was in accordance with the results of our experiment performed in the suspension cultured *C. paliurus* cells. Their data also showed that NO quenching suppressed H₂O₂ synthesis, and exogenous NO addition increased H₂O₂ accumulation. According to our results, a change in NO concentration had no effect on H₂O₂ synthesis. Halder et al. (2019) showed that NO synthesis occurred downstream of the ROS burst under elicitation, which is in accordance with our results. These phenomena be caused by plant species differences or other unknown reasons.

Both SA and JA are important signal molecules associated with plant defense responses (Kunkel and Brooks et al. 2002). Our previous studies showed that any one of SA or JA induction could stimulate the synthesis of secondary metabolites (Liu et al. 2018b). However, according to Li et al. (2004), there was antagonism between SA and JA pathways in *A. thaliana* in response to elicitation, and WRKY70 acted as an activator of SA-induced genes and a repressor of JA-responsive genes to integrate signals from these mutually antagonistic pathways. Similar results were reported by Xu et al. (2006b). In our ANE elicitation

experiments, JA concentration increased significantly after ANE addition, and played a critical role in stimulating triterpenoids synthesis, while SA concentration was lower than that of the untreated cells. Our results also confirmed the antagonism between SA and JA pathways.

In this study, JA was the critical point in the entire signal network, which connected with NO and H₂O₂ signal pathways. Previous studies indicated that JA, as an important endogenous phytohormone, was synthesized using α -linolenic acid as starting compound (Liechti et al. 2006), and a series of cascade enzymes were involved, including LOX, AOS, AOC, and OPR (Mueller et al. 1997; Toyoda et al. 2013), which were also identified in our elicited *C. paliurus* cells (Table 1). Our RNA-seq analysis showed that a total of three candidate JA synthesis pathway genes including one LOX and two OPR genes were significantly up-regulated under ANE stimulation, as well as five down-regulated JAZs and one up-regulated JAR1 relating to JA signal transduction, which further validated the key role of JA in response to ANE elicitation. JAs (jasmonates) usually refer to oxygenated compounds known as oxylipins and derived from JA, including jasmonic acid (JA), methyl jasmonate (MeJA), and jasmonoyl-isoleucine (JA-Ile), which act as ubiquitous and conserved regulators for the production of secondary metabolites in the majority of plants, from gymnosperms to angiosperms (Afrin et al. 2015; Avanci et al. 2010). It is generally recognized that JA shows its effects through a bioactive form—JA-Ile, which is synthesized from JA and isoleucine under the catalysis of JA amino acid synthetase (JAR1) (Staswick and Tiriyaki et al. 2004; Afrin et al. 2015). Normally, JA-Ile concentration is very low, and MYC2 is bound by jasmonate ZIM-domain protein (JAZ) and presents in an inactive form. When the JA-Ile concentration increases to a certain level under stimulation, the COI1 protein of the SCF^{COI1} complex interacts physically with JAZ protein labels it with ubiquitin, which causes MYC2 to be freed from JAZ and JAZ to be degraded by 26S-proteasome. The free MYC2 released from JAZ activates the expression of the early jasmonate-response genes and finally improves the synthesis of secondary metabolites to adapt to environmental changes (Major et al. 2017; Afrin et al. 2015; Avanci et al. 2010; Ge et al. 2015; Commault et al. 2019). Our results also indicated that JA mediated the signal transduction of ANE elicitation, in which transcription factors like JAZ and JAR play important roles. Meanwhile, MYC2 was also annotated in our ANE treated cells, but its expression declined slightly at the 20 h after elicitation (Table 2). However, further studies are still needed to interpret the underlying mechanism of these transcription factors associated with JA signal transduction pathways.

Conclusion

ANE elicitation significantly enhanced triterpenoids accumulation in suspension cultured *C. paliurus* cells. NO, H₂O₂ and JA were all involved in this elicitation as important signal molecules. Three related signal pathways were deduced based on a series of experiments involving the blocking of signal molecules, together with three hypothetical signal transduction pathways involved in ANE elicitation. JA was the critical point in the signal network of ANE elicitation, which connected with NO and H₂O₂ signal pathways. RNA-seq analysis further validated the pivotal role of JA in ANE elicitation.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s13562-021-00764-z>.

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Authors' contributions ZY and JC conceived the idea and designed the experiments. MW and QX performed the experiments. MW, XX and XZ contributed to data analysis. All authors discussed the results and commented on the manuscript. All authors read and approved the final manuscript.

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Data availability All relevant data are within the paper and its Supporting Information files.

Declarations

Conflict of interest The authors declare no competing financial interests.

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